Stanley Nelavala

Software Engineer: AI-First Product and Platform (TypeScript, React Native, Postgres, AWS)

218-213-8220 | consultantstanleymn@gmail.com | LinkedIn | GitHub | stanley-n.com
Huntington, IN | Open to relocate anywhere

Summary

Software engineer who ships AI-powered features and the reliable platform under them. Public work: a LangGraph and MCP agent platform (human approval, audit log, 62 evals, LLM failover) and a Java payments ledger with idempotency keys and SLOs. At a 10-person hospital startup I own features from design to release across Python APIs, React and React Native. I use Claude Code, Codex and opencode daily to design, write, debug and review code, and I check agent output with tests and acceptance checks. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Search pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant (no cloud call), and RxDesign, the hospital design service (FastAPI, React, TypeScript; 220+ offline tests).
- Set up separate iOS and Android release pipelines in GitHub Actions with automatic TestFlight submission.
- Use Claude Code, Codex and opencode every day to design, write, debug and review code; tests and acceptance checks gate every change.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session.

Projects (public code, tests)

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis, Terraform* GitHub

- AI case workflow with typed state, human approval that resumes after restart, step and token budgets and one constrained MCP write tool; 62 deterministic replay evals, prompt-injection tests, PII redaction, an append-only audit log, and failover between two LLM adapters with a circuit breaker.

Payments Ledger Service | *Java 25, Spring Boot, PostgreSQL, Kafka API, React, TypeScript, Prometheus, k6* GitHub

- Double-entry ledger (authorize, capture, void, refund, transfer) with idempotency keys and a transactional outbox; property tests and a k6 check that no money is created or lost; 28-day error budgets; React and TypeScript ops console tested with Playwright and axe.

Inference Gateway | *Go, gRPC, ONNX Runtime, PyTorch, OpenTelemetry, Helm* GitHub

- Go service for an int8-quantized classifier with micro-batching, deadlines and load shedding; a Little's law model sizes a 10x launch peak at 2 to 3 replicas (local benchmark); Helm chart tested on kind.

Agent-fleet build workflow | *Claude Code, opencode, DeepSeek, shell* GitHub

- A planner, manager and worker pipeline that delivered two public services and an ops console in 34 work packets, each with written acceptance checks, a progress log and a manager check.

Skills

AI-first development: Claude Code and subagents, Codex, opencode, a DeepSeek worker harness, Claude in Chrome, Gemini, OpenRouter models; agent runs with acceptance checks
AI features: LangGraph, MCP, retrieval (pgvector), deterministic evals, guardrails, PII redaction, LLM failover
Languages: TypeScript, JavaScript, Python, Java, Go, Rust, C#, SQL, Kotlin (local project)
Product and mobile: React, Redux-style state, React Native (Expo), REST, gRPC, FastAPI, Spring Boot
Platform: PostgreSQL, Redis, AWS (ECS, RDS, S3, SQS), Terraform, Docker, Helm on kind, GitHub Actions, OpenTelemetry, Prometheus, SLOs

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)